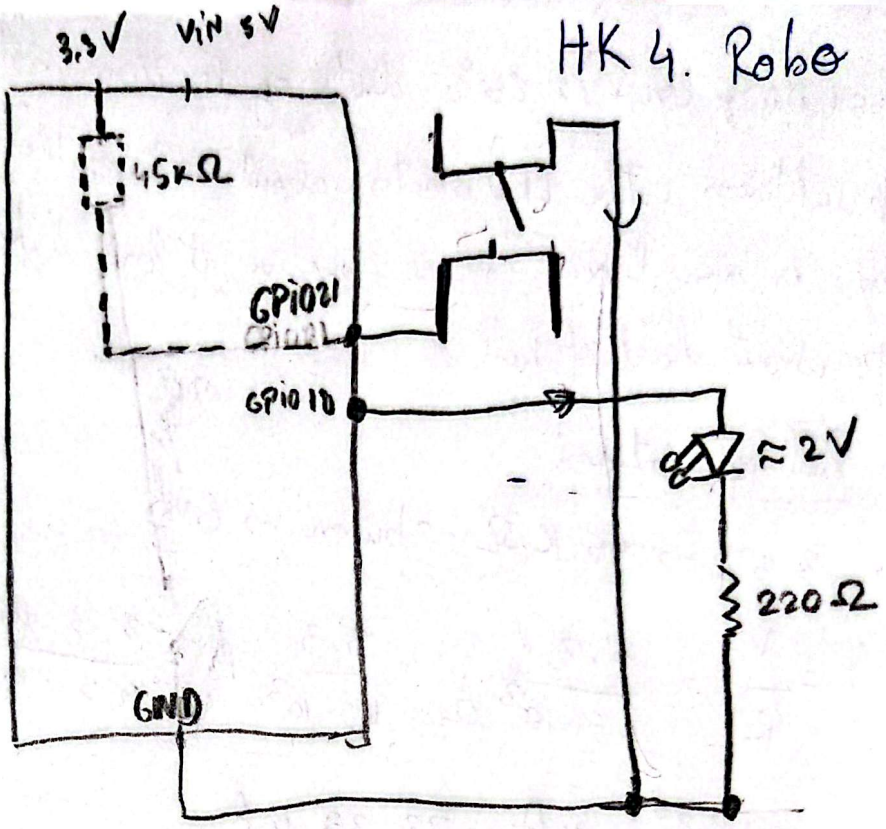
HK 4. Robo

INTERNAL PULL UP



1. Curent prin LED

$$I = \frac{V}{R} = \frac{V_{PIN} - V_{LED}}{R} = \frac{3.3V - 2V\,(led\ nou)}{220\,\Omega} = \frac{1.3\,V}{220\,\Omega} = \frac{1.3}{220}\,A =$$

$$= \frac{1300}{220}\,mA = 5,909\,mA \leq 12mA \quad (nu\ ardem\ pin\text{-}ul)$$
$$\approx 5,91\,mA$$

2. Calcul MOD DIM (PWM)

Aflăm ce primește LED-ul când scriem analog Write (18,64)

Duty cycle: $D = \frac{T_{on}}{T} = \frac{\text{cât timp stă pin-ul pe HIGH într-o perioadă}}{\text{perioada totală a semnalului PWM (ON+OFF)}}$

$$D = \frac{PWM\ value}{PWM\ max} = \frac{64}{255}\ (8\ biți) = 0,2509 \approx 0,251 = 25,1\,\%$$

Tensiune medie $\Rightarrow V_{avg} = D \cdot V_{HIGH} = 0.251 \cdot 3.3V = 0,8283\,V \approx 0.83V$

Curent med. prin LED $\Rightarrow i_{avg} = D \cdot i_{ON} = 0.251 \cdot 5,91mA = 1,48341\,mA \approx$
$1,48mA$

1/

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Obs (Duty Cycle): 25% duty cycle does not mean 25% perceived brightness. The LED emits about 25% of the light but the human eye is non linear (more sensitive at low light), so DIM looks closer to about half of full.

- Pull-up intern

3.3V → 45 KΩ → buton → GND.

$$I = \frac{V}{R} = \frac{3.3\,V}{45 \cdot 10^3\,\Omega} = \frac{3.3}{45 \cdot 10^3}\,A = \frac{3.3 \cdot 10^3}{45 \cdot 10^3}\,mA = \frac{3.3 \cdot 10^3}{45}\,\mu A$$

$$= \frac{3300}{45}\,\mu A = 73,33\,\mu A.$$

$$P = \frac{V^2}{R} = \frac{10.89\,V}{45 \cdot 10^3\,\Omega} = \frac{10890 \cdot 10^3}{45 \cdot 10^3}\,\mu W = \frac{10890}{45} = 242\,\mu W$$

Obs: button released → open circuit → $I \approx 0$ → no current flows, no voltage drop on the pull-up resistor so GPIO 21 sits at 3,3 V (High)

button pressed → loop closes → small current 73,33 μA flows

GPIO is pulled to 0 V (Low), but the large resistance 45 KΩ keeps the current small so no power is wasted.

- Vom calc. perioada și frecvența.

$T = t_{ON} + t_{OFF}$ (ciclu complet)

$\nu = \frac{1}{T}$ (câte cicluri pe sec)

$1\,ms = 1 \cdot 10^3\,ms$

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Mărime.	Date	Inc.	T(s)	ν(Hz)
Debounce	30 ms	0,03 s	–	–
Blink (lampă)	500 ms ON + 500 ms OFF	–	~~1 s~~	~~1~~ 1 Hz
Heartbeat (GPIO2)	Comută la 500 ms	–	0,5 s	1 Hz
Long press	1000 ms	1 s	–	–

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